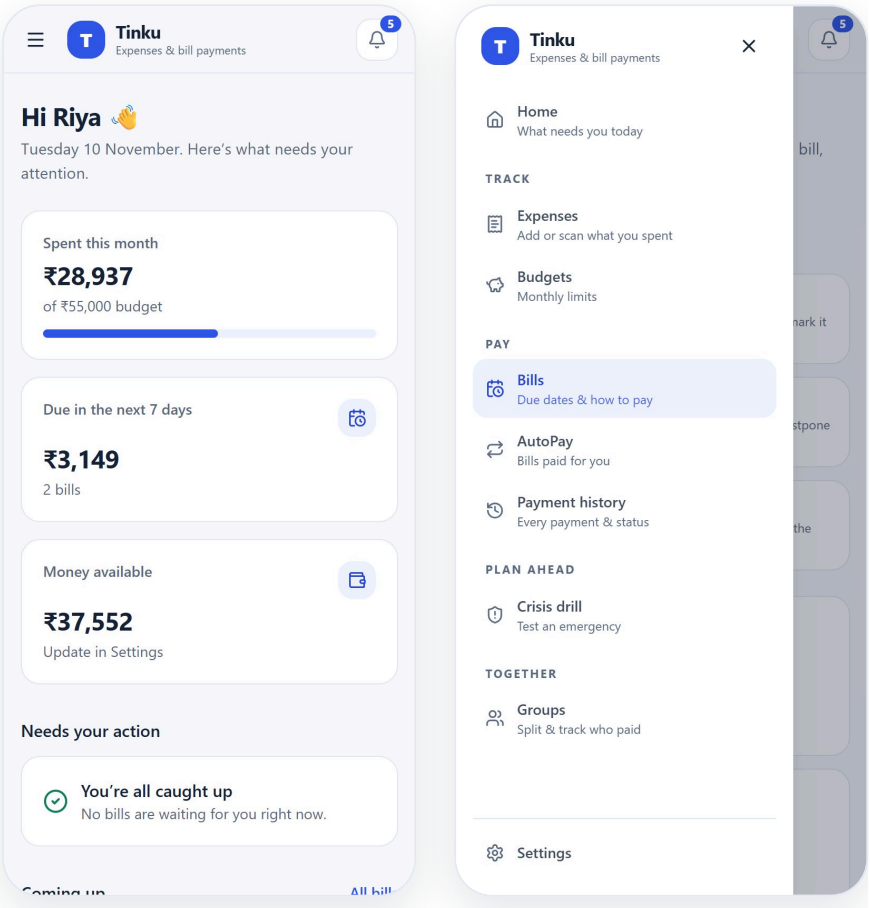
Tinku
Smart expenses & bill payments

Feature flows, page by page

How every screen of Tinku works: what you see, what you do, and what happens behind the scenes. One section per page, in the same order as the app's menu.

01	App map & ground rules	p. 2
02	Home	p. 3
03	Expenses & receipt scan	p. 4
04	Budgets	p. 5
05	Bills & payment modes	p. 6
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13	End-to-end journeys & statuses	p. 17



Project
PayPal AI hackathon · team build

Version
MVP, 10 November 2026

Screens
Captured from the running app with demo data

0 App map & ground rules

Every feature has its own page. The left menu groups them by job.

Tinku helps one budget owner **track** spending, **plan** bills, **control** how each bill is paid, **share** costs with others, and **rehearse** money emergencies before they happen.

Home
What needs you today · /

TRACK

- **Expenses** `/expenses`
  - Add expense + receipt scan
- **Budgets** `/budgets`
  - Edit limits

PAY

- **Bills** `/bills`
  - Add · edit · details
  - Pay: review → checkout → result
- **AutoPay** `/autopay`
  - Authorize with a limit
- **Payment history** `/history`

PLAN AHEAD

- **Crisis drill** `/drill`
  - Pick emergency → result → recovery plan → save

TOGETHER

- **Groups** `/groups`
  - New group
  - Split · settle · remind

Always available: Notifications (bell in the top bar, `/notifications`) · Settings (bottom of the menu, `/settings`) · the demo badge in the top bar showing “Demo payments” or “PayPal Sandbox”. Full routes for every page are in the appendix.

Ground rules that apply on every page

One page, one job
Each page header says what the page is for and shows its main action as the big blue button.

A bill is paid only when verified
Never because a button was clicked or a popup closed. The server confirms the provider’s result (status, amount, currency, reference) first.

Nothing is charged twice
Every payment carries an idempotency key; repeated clicks, retries and duplicate webhooks are ignored.

Essential comes first
Rent, utilities, fees and health are marked essential. Emergency plans protect them and trim flexible spending instead.

Honest demo labels
PayPal India can’t take domestic INR payments, so the PayPal step is a Sandbox (USD) demo or a clearly labelled simulator. Real Indian bill settlement is future scope.

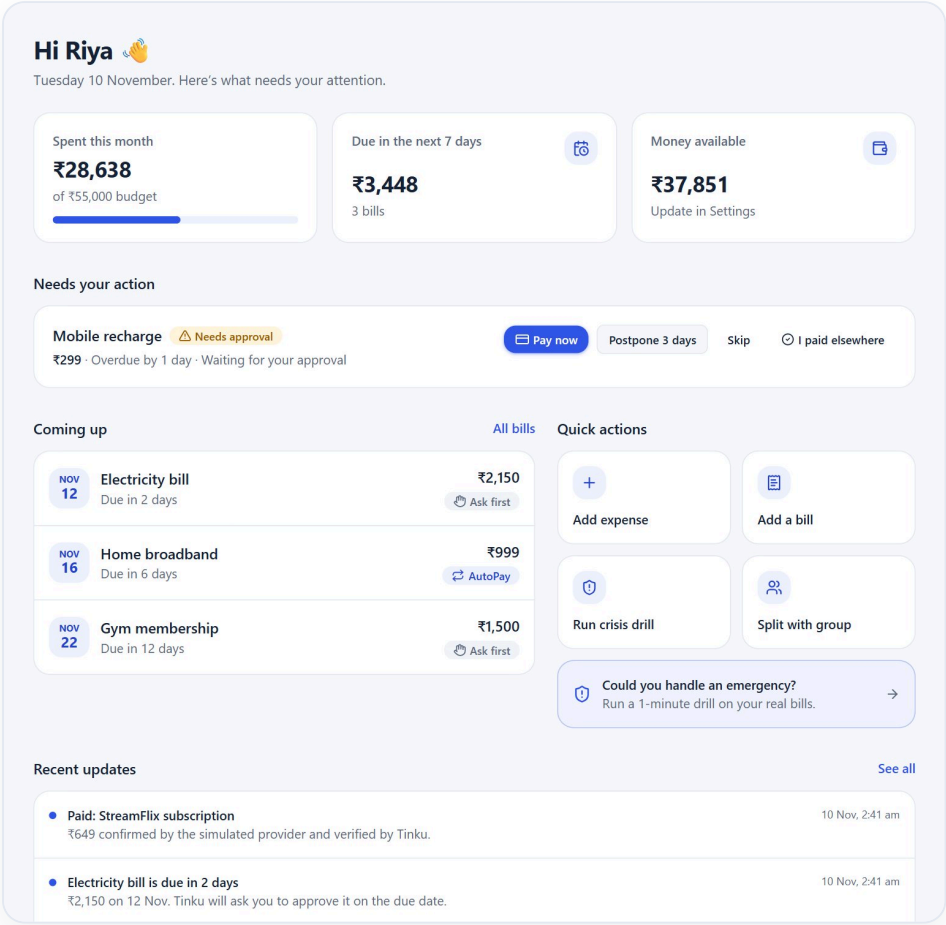
Everything leaves a trail
Every action creates a notification, and every payment keeps a step-by-step timeline.

Daily check
Once a day (on the first visit, or by a cron call) Tinku sends reminders, marks bills due, and runs AutoPay.

1 Home

Menu → Home

Your daily starting point. It shows **only what needs you today**, so you know what to do the moment you open the app.



Home: three summary tiles, the bill waiting for approval, what's coming up, and quick actions.

Flow

1. **Open Tinku**
   The daily check runs once per day: reminders 3 days ahead, due bills, AutoPay.
2. **Read the three tiles**
   Spent this month vs budget · due in the next 7 days · money available now.
3. **Handle "Needs your action"**
   Each bill shows why it needs you. Act right there: Pay now, Postpone 3 days, Skip, or I paid elsewhere.
4. **Check "Coming up"**
   The next 14 days with each bill's payment mode. Tap a bill to open it.
5. **Use a quick action**
   Add expense · Add a bill · Run crisis drill · Split with group.

Behind the scenes

- Tiles are calculated live from this month's expenses, budgets and open bills.
- "Money available" goes down automatically when you record an expense or a bill is paid.
- The buttons shown depend on the bill's payment mode (see Bills).
- Recent updates lists the latest notifications; unread ones have a blue dot.

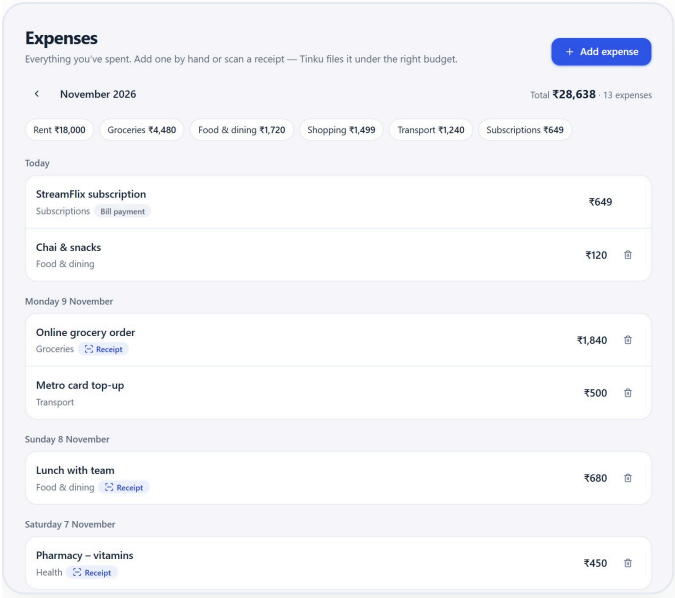
Empty state: when nothing is waiting, Home shows "You're all caught up".

2 Expenses & receipt scan

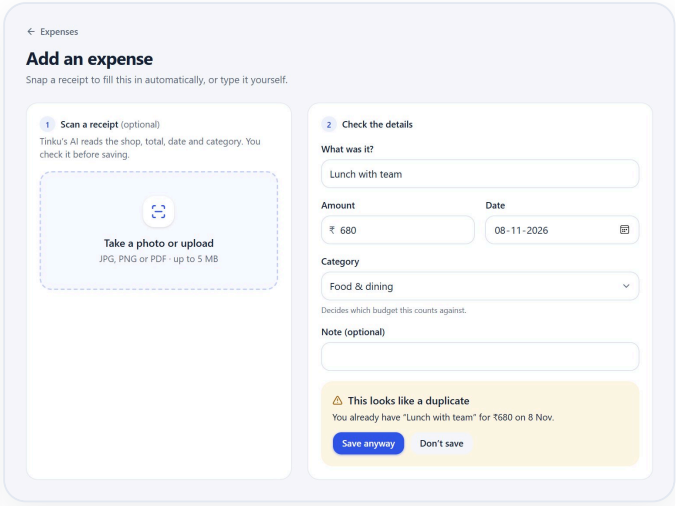
Menu → Track → Expenses

`/expenses` `/expenses/new`

Record what you spent, **by hand or from a photo of the receipt**. Every expense counts against a budget.



List by month, with totals and top categories.



Add expense: receipt scan + check the details. A duplicate asks first.

Flow: add an expense

1. **Tap "Add expense"**
   From Expenses, Home, or the quick actions.
2. **(Optional) Scan a receipt**
   Take a photo or upload JPG / PNG / PDF up to 5 MB.
3. **AI fills the form**
   Claude reads the shop, final total, date and category. The form is marked "please check", and low confidence or a foreign currency is called out.
4. **Check the details**
   What it was · amount · date · category (decides the budget) · note.
5. **Save expense**
   Tinku looks for a duplicate: same amount, within one day, same category or name.

No duplicate
Saved, the budget updates, money available goes down, and you return to the list.

Looks like a duplicate
Shows the existing entry. Choose "Save anyway" or "Don't save".

On the list

- Move between months with ← →; the current month is the default.
- Grouped by day, with "Receipt" and "Bill payment" tags.
- Delete manual or receipt entries (the money goes back to available).

Edge cases handled

Blurry or not a receipt	Clear message; type it in by hand.
No AI key set	Scanning says it isn't set up; manual entry always works.
AI misreads	Nothing is saved until you press Save.
Bill-payment expense	Can't be deleted; it stays for your records.

3 Budgets

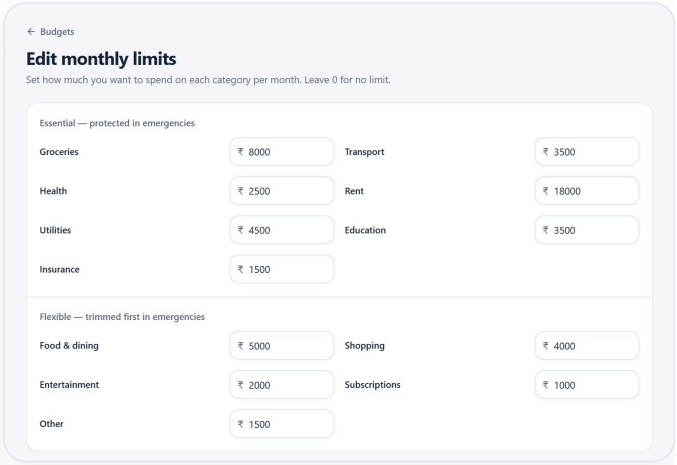
Menu → Track → Budgets

`/budgets` `/budgets/edit`

A **monthly limit for each kind of spending**, so you can see at a glance where you stand.



Overall bar, then one bar per category.



Edit limits, grouped Essential / Flexible.

Flow

1. **Open Budgets**
   Total spent of total budget, money left and days to go this month.
2. **Scan the bars**
   Blue is fine, amber means 80% used, red means over. The text always says “₹X left” or “₹X over”.
3. **See the warning if over**
   Lists the categories over budget and suggests a Crisis drill.
4. **Edit limits**
   Set ₹ per category (0 = no limit) → Save limits.

Behind the scenes

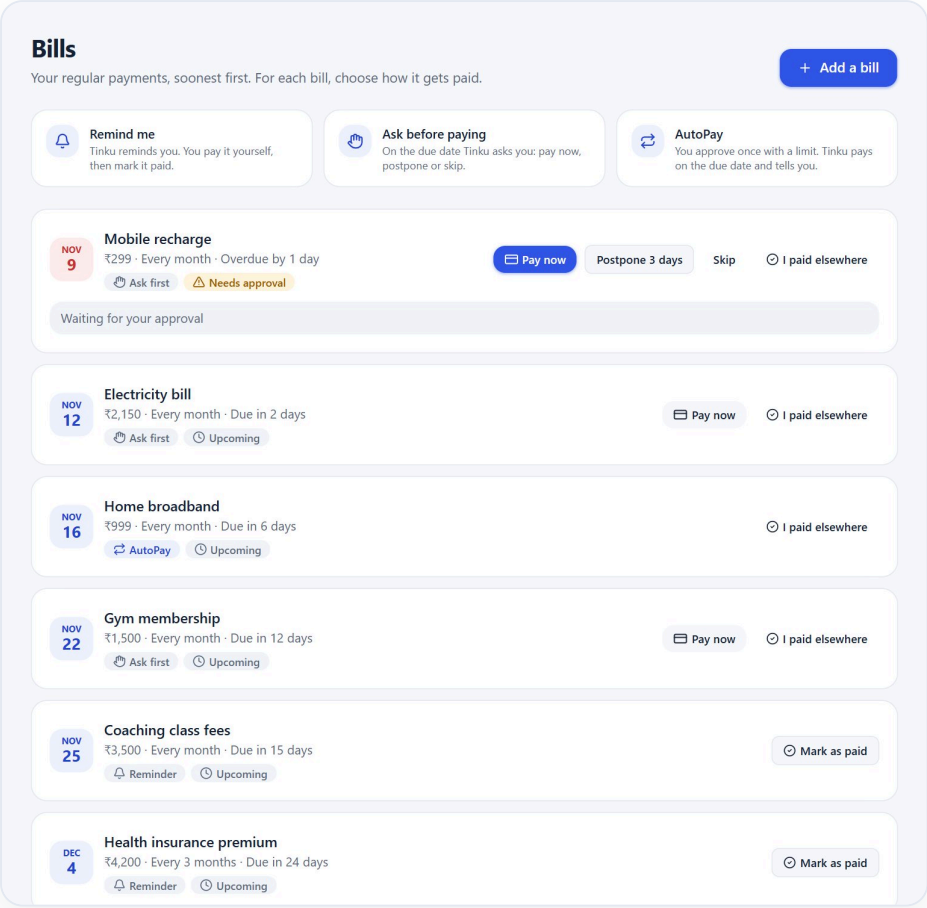
- Spending comes from Expenses, including bills paid through Tinku or marked paid.
- **Essential** (groceries, transport, health, rent, utilities, education, insurance) vs **Flexible** (food & dining, shopping, entertainment, subscriptions, other).
- The Crisis drill uses these tags to decide what to trim first in an emergency.

4 Bills & payment modes

Menu → Pay → Bills

`/bills` `/bills/new` `/bills/[id]` `/bills/[id]/edit`

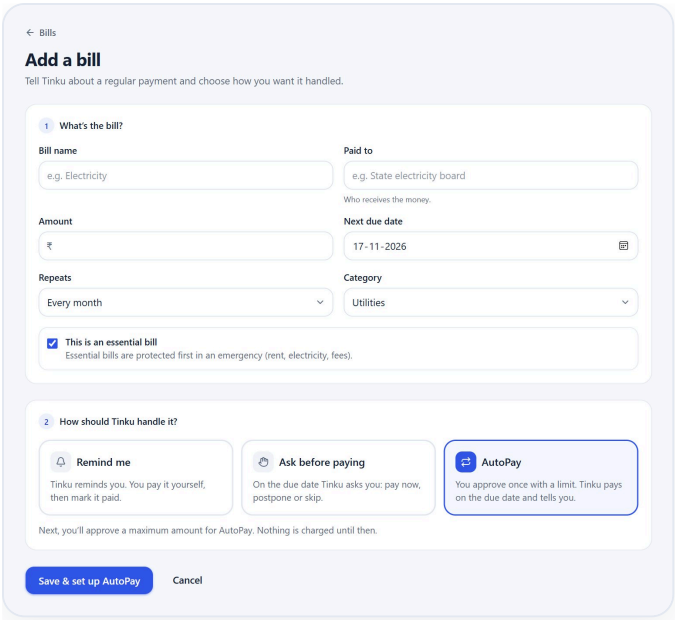
Your regular payments, soonest first, and **your choice of how each one gets paid**.



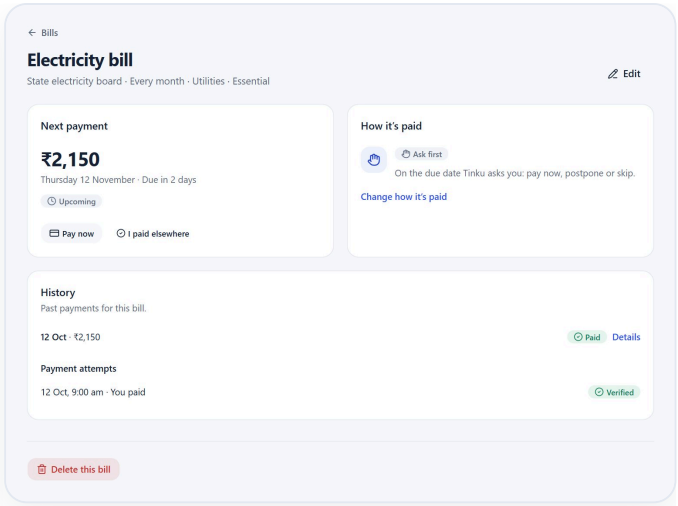
Bills: the three modes explained at the top, then each bill with its date, mode, status and the right buttons.

The three payment modes (chosen per bill)

Mode	What happens on the due date	Good for
Remind me	A reminder 3 days before and on the day. You pay it your usual way, then tap “Mark as paid”. Tinku never pays.	Rent, school fees
Ask before paying	The bill turns “Needs approval”. You choose Pay now, Postpone 3 days or Skip.	Electricity, phone
AutoPay	Paid automatically, only within the limit you authorized. If anything is off, it asks you instead.	Subscriptions, broadband



Add a bill: details, then pick a mode.



Bill details: next payment, how it's paid, history.

Flow: add a bill

1. **Tap "Add a bill"**
   Name, paid to, amount, next due date, repeats (once / weekly / monthly / every 3 months / yearly), category.
2. **Essential?**
   Pre-ticked from the category; essential bills are protected first in emergencies.
3. **Choose the mode**
   Three big cards: Remind me · Ask before paying · AutoPay.
4. **Save**
   AutoPay goes on to "Authorize AutoPay" (nothing is charged until you do). Other modes open the bill page.

Bill details page

- Next payment: amount, date, "postponed from …", status, and why it needs you.
- How it's paid, plus the AutoPay limit and expiry when on AutoPay.
- History of past payments and every attempt, with links to their timelines.
- Edit or delete the bill (history stays; AutoPay is cancelled).

Which buttons appear

Status	Buttons
Upcoming **Ask before paying**	Pay now (early) · I paid elsewhere
Upcoming **Remind me**	Mark as paid
Due **Remind me**	Mark as paid · Postpone 3 days · Skip
Needs approval Failed **Ask first or AutoPay**	Pay now / Try again · Postpone · Skip · I paid elsewhere
Processing	Check payment status
AutoPay, not authorized	Authorize AutoPay

Edge cases handled

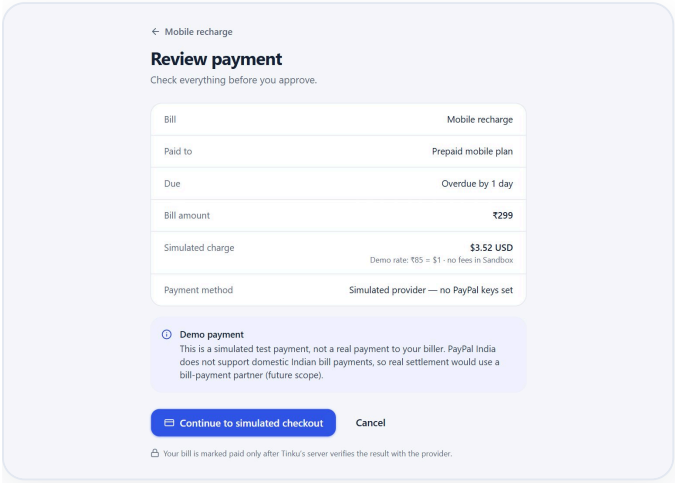
Postpone	Moves only this payment; the bill still repeats as usual.
Skip	This one stays unpaid; the next one is not affected.
Paid outside the app	Mark as paid. Blocked while a payment is processing.
Overdue	The date tile turns red and the label says "Overdue by N days".

5 Paying a bill

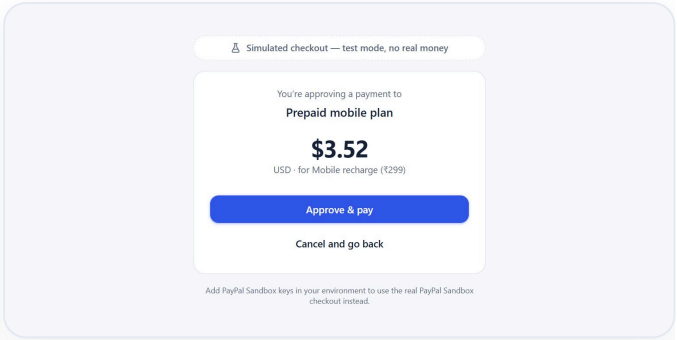
"Pay now" on Home, Bills or a bill page

`/pay/[id]` `PayPal or /pay/simulated` `/payments/[id]`

Pay one bill with a **clear review step** and a **verified result**.



1 · Review: rupee amount, USD Sandbox charge, method, demo notice.



2 · Approve: PayPal Sandbox, or this labelled simulator when no keys are set.

Flow

1. **Tap "Pay now"**
   Opens the review page for that bill.
2. **Review**
   Bill, paid to, due, amount in ₹, the USD charge at the demo rate ₹85 = $1, payment method.
3. **Continue**
   The server creates the payment (and a PayPal order when keys are set) with an idempotency key.
4. **Approve**
   On the PayPal Sandbox page, or "Approve & pay" in the simulator. Cancel returns with nothing charged.
5. **Server captures & verifies**
   Checks the status is COMPLETED and the amount, currency and our reference all match.
6. **Result page**
   Shows what happened, with a full timeline.

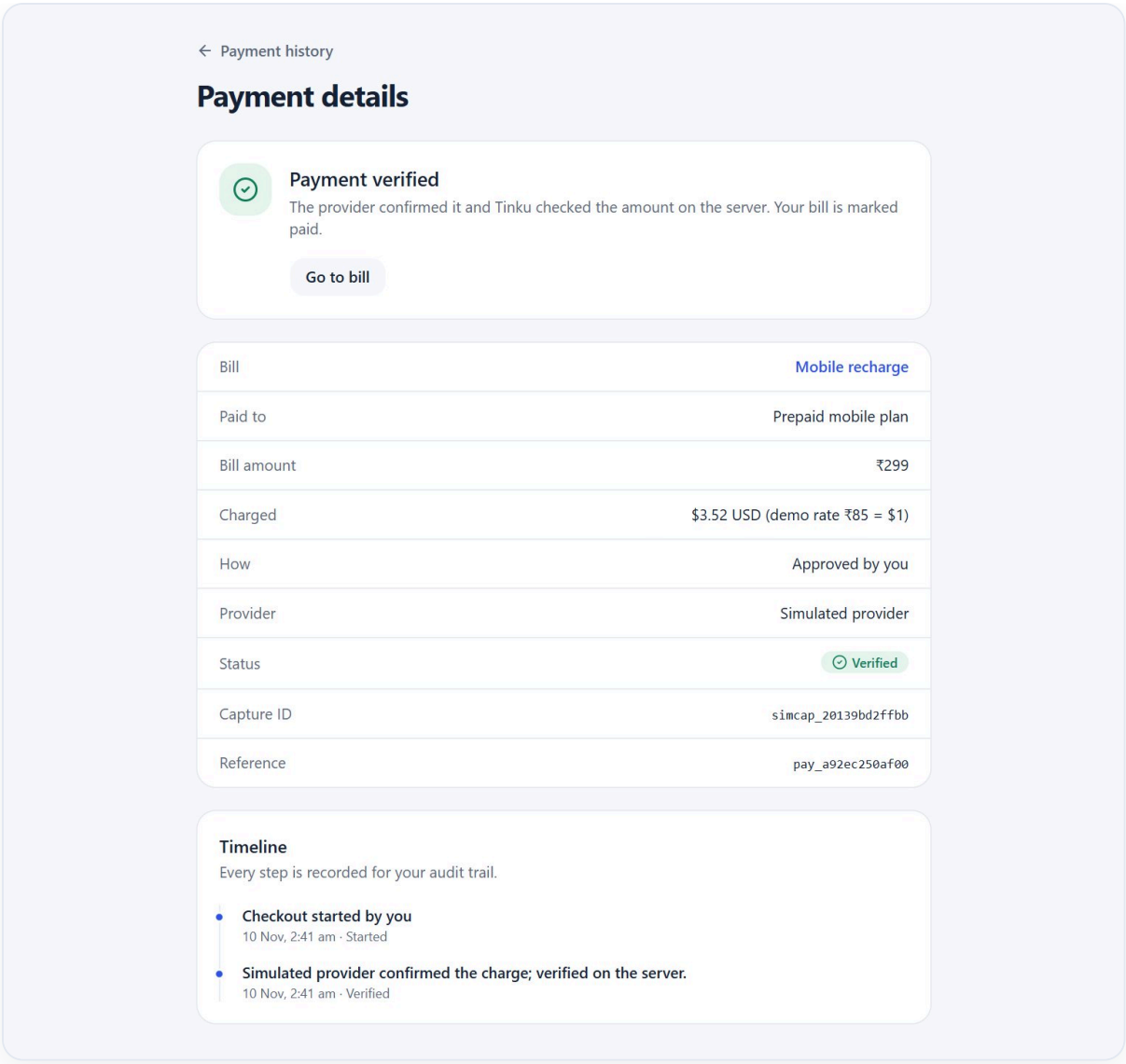
Possible results

Verified
Bill marked paid, expense recorded, money available lowered, next due date opened, notification sent.

Pending
Bill shows "Processing". "Check status again" asks PayPal; nothing is ever charged twice.

Failed
Nothing is marked paid; the bill stays unpaid. "Try again" starts a fresh checkout.

Cancelled
You left the checkout. Nothing charged, bill unchanged.



3 · Result: “Payment verified”, the details, and a timeline that serves as the audit trail.

Behind the scenes (safety)

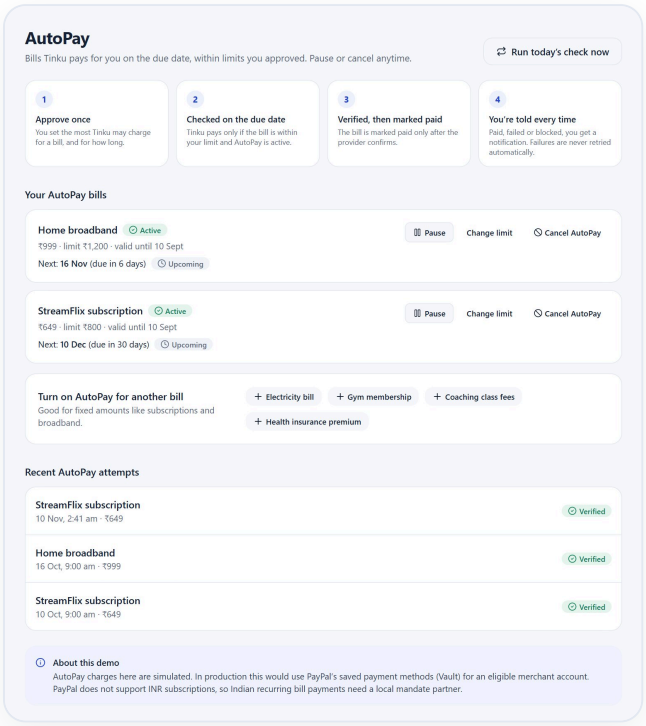
Secrets stay on the server	The PayPal client secret never reaches the browser; orders are created and captured server-side.
Idempotency	PayPal-Request-Id on create and capture. An abandoned checkout is replaced, so only one can ever complete.
Webhooks	PAYMENT.CAPTURE.* events are signature-checked with PayPal, de-duplicated by event id, and must match the amount.
Final means final	Once a payment is verified, a late “failed” event can’t undo it.
Honest scope	Demo charges are Sandbox USD. Real settlement with Indian billers needs a BBPS partner (future scope).

6 AutoPay

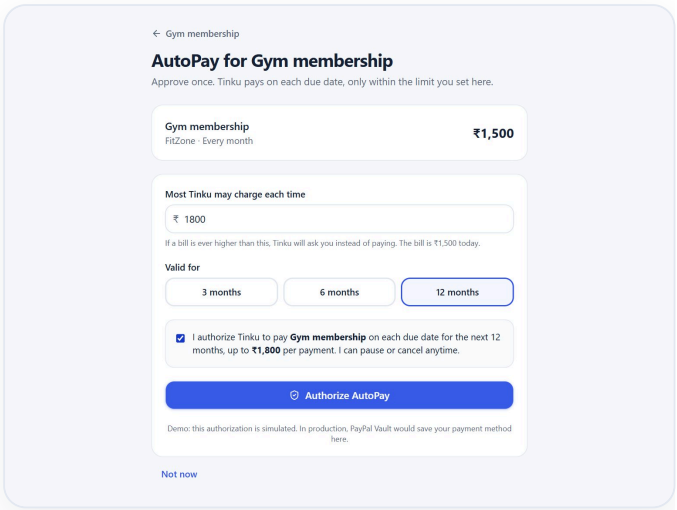
Menu → Pay → AutoPay

`/autopay` `/autopay/new?bill=…`

Bills Tinku pays for you, **only within limits you approved**. Pause or cancel anytime.



How it works in 4 steps, your AutoPay bills, recent attempts.



Authorize once: limit, duration, explicit consent.

Flow: turn on AutoPay

1. **Start from a bill**
   Choose AutoPay when adding the bill, or "Turn on AutoPay for another bill".
2. **Set the limit**
   The most Tinku may charge each time (suggested: bill + 20%, never below the bill).
3. **Pick how long**
   3, 6 or 12 months.
4. **Tick consent**
   "I authorize Tinku to pay … up to ₹X per payment." The button stays disabled until it's ticked.
5. **Authorize**
   The bill switches to AutoPay and you get a notification.

Manage

- Pause / Resume · Change limit · Cancel AutoPay (the bill goes back to "Ask before paying").
- "Run today's check now" runs the daily check on demand (great for demos).

On the due date: the daily check

Authorization exists?	No → Needs you
Still valid (not expired)?	No → re-authorize
Active (not paused)?	No → Needs you
Amount ≤ your limit?	No → approve this one yourself
Not already attempted?	Already tried → skip (no double charge)
Charge	Verified paid + notified · Failed stays unpaid

Never auto-retried: a failed AutoPay charge waits for you, as the product spec requires.

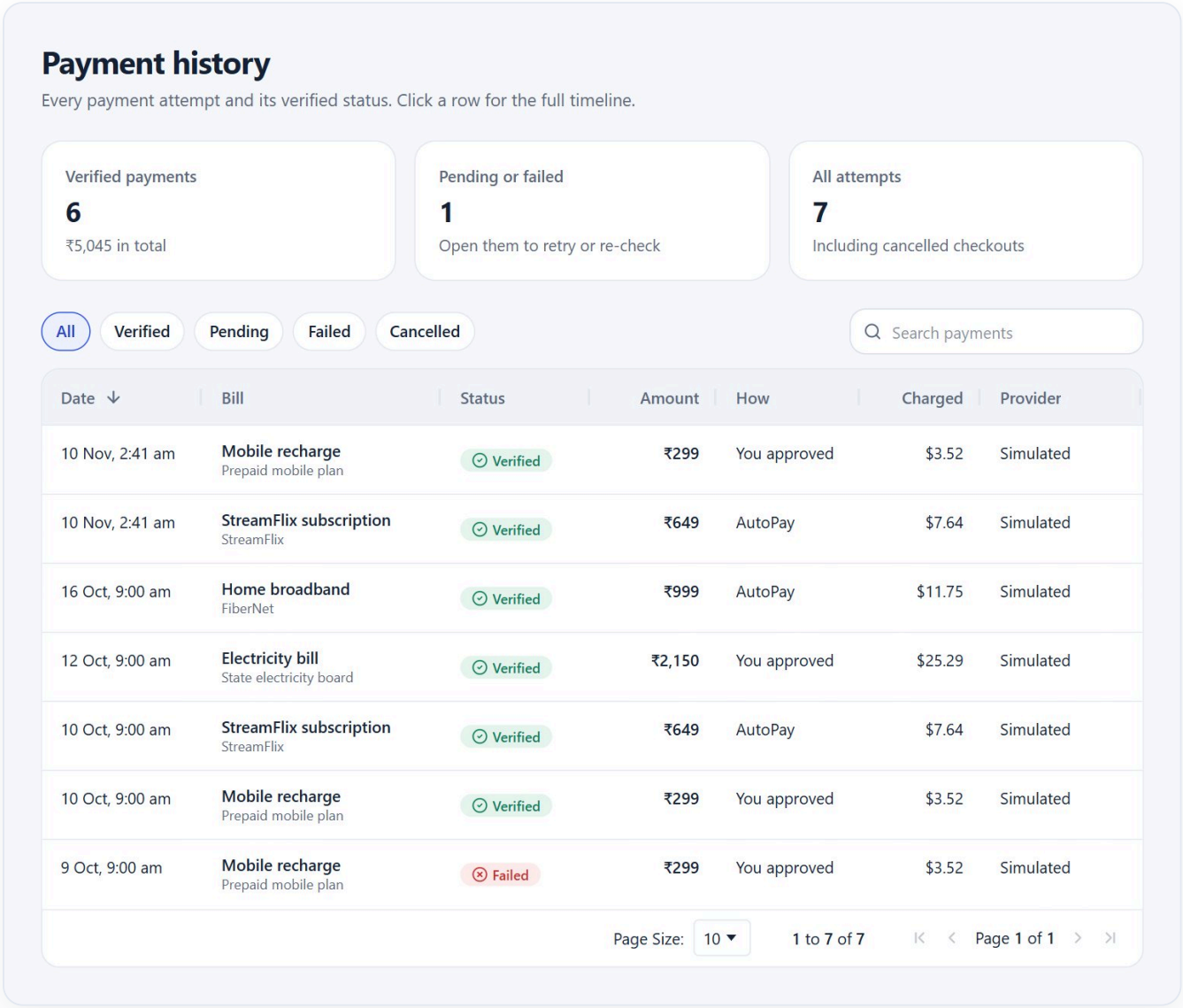
Demo note: AutoPay charges are simulated. Production would use PayPal Vault on an eligible merchant; PayPal doesn't support INR subscriptions.

7 Payment history

Menu → Pay → Payment history

`/history`

Every payment attempt and its **verified status**: your audit trail in one table.



Summary tiles, status filters, search, and a sortable table (AG Grid). Click a row for its timeline.

Flow

1. **Read the tiles**
   Verified payments and total · pending or failed · all attempts.
2. **Filter or search**
   All · Verified · Pending · Failed · Cancelled, plus free-text search.
3. **Sort**
   Click any column header; newest first by default.
4. **Open a row**
   Goes to the payment page with its full step-by-step timeline.

Columns

- Date · Bill (and payee) · Status · Amount (₹)
- How: AutoPay or You approved
- Charged (USD demo amount) · Provider (PayPal Sandbox or Simulated)

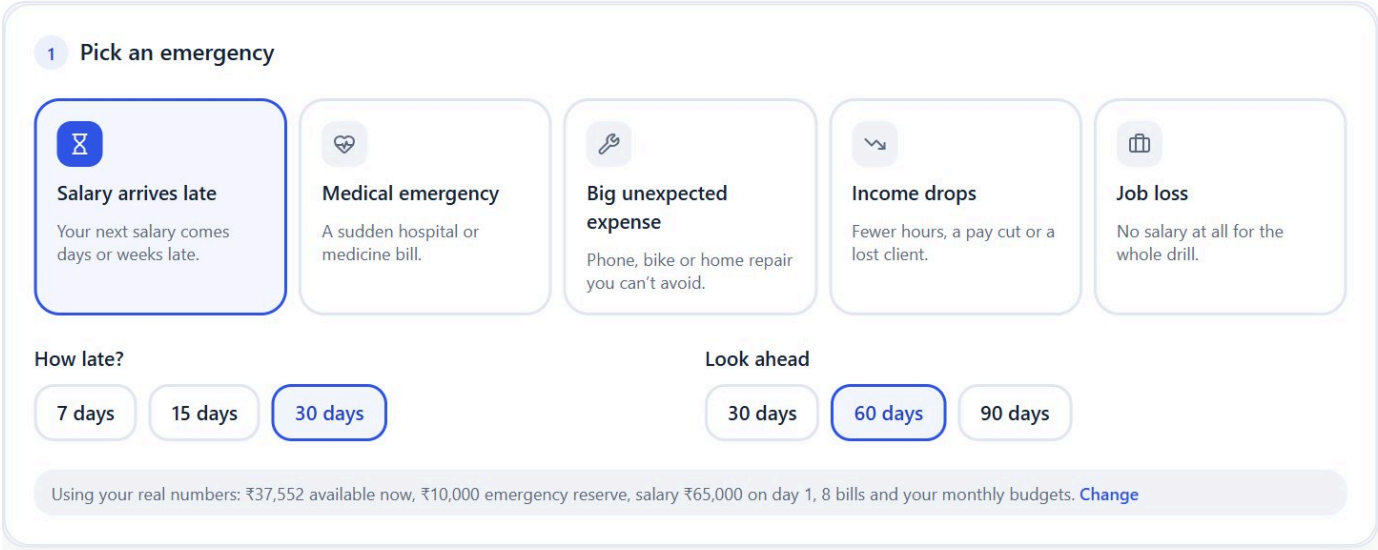
Built with **AG Grid Community**: sorting, quick search and pagination.

8 Crisis drill

Menu → Plan ahead → Crisis drill

`/drill`

A **fire drill for your money**. Test an emergency on your real bills and budget before it ever happens, and get a recovery plan.



Step 1: pick an emergency and its size, and how far to look ahead.

Flow

1. **Pick an emergency**
   Salary arrives late · Medical emergency · Big unexpected expense · Income drops · Job loss.
2. **Set the details**
   Days late (7 / 15 / 30), amount (₹5k–₹50k or custom), % drop (20 / 30 / 50); look ahead 30 / 60 / 90 days.
3. **Tinku simulates every day**
   Uses your real balance, salary day, bills and budgets. Essential bills are paid first.
4. **Read the verdict**
   "You'd get through" or "You'd run short by ₹X on date", a resilience score out of 100, a day-by-day chart, and the bills that would bounce.
5. **Use the recovery plan**
   Least disruptive first. Tick or untick steps; the chart and result update instantly.
6. **Save this drill**
   Kept in "Past drills" with a notification. "Ask your group" opens Groups to collect help.

The recovery plan, in order

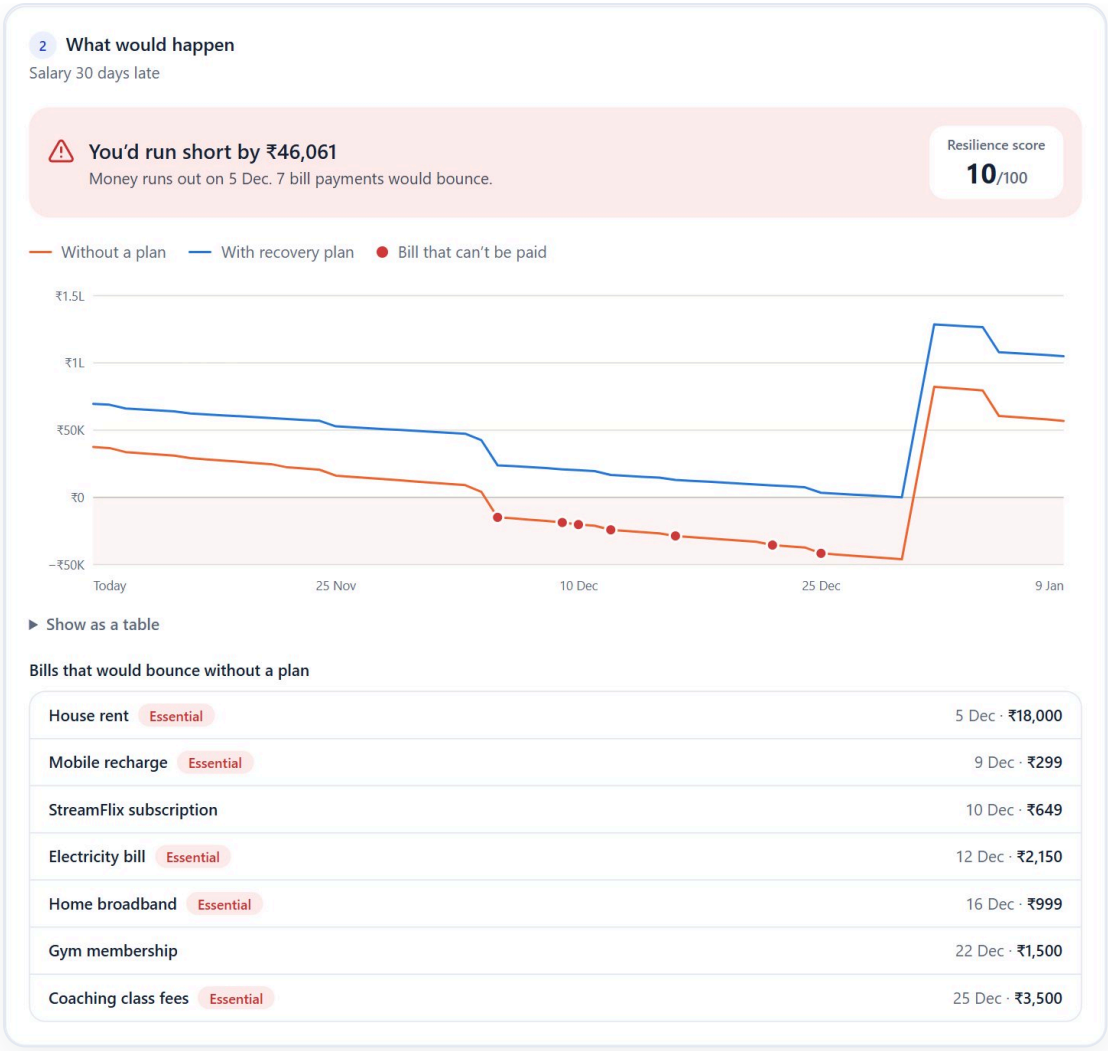
1 · Use your emergency reserve
Savings move into spending money.

2 · Halve flexible spending
Food & dining, shopping, entertainment… for the drill period.

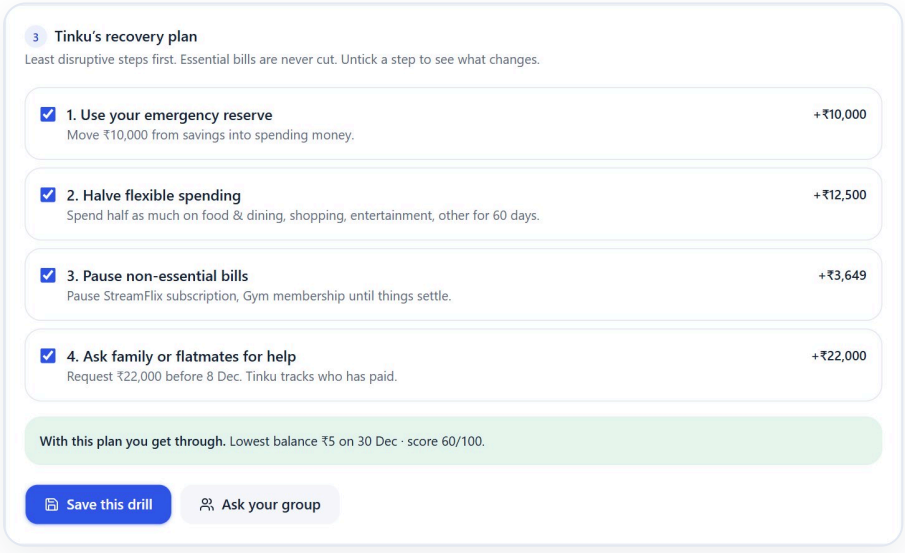
3 · Pause non-essential bills
e.g. subscriptions and the gym, until things settle.

4 · Ask family or flatmates
Requests exactly the remaining gap, before the date money runs out.

Score: 80+ strong · 60–79 okay · below 60 runs short. Essential bills are never cut.



Step 2: the verdict, the balance chart (orange = without a plan, blue = with the plan, red dots = bills that bounce), and the bills at risk.



Step 3: the recovery plan. Here a 30-day salary delay goes from ₹46,061 short to getting through.

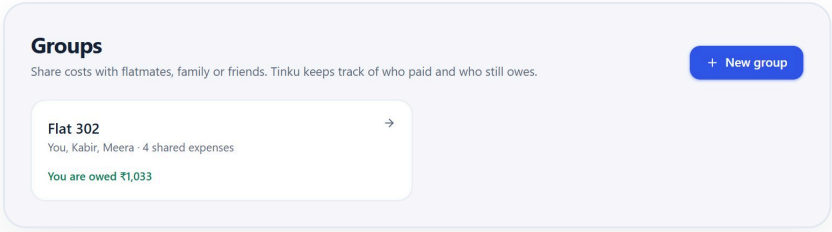
Why it's trustworthy: the rupee maths is plain, deterministic code (not AI), so every number is exact and explainable. Saved drills are recalculated on the server from your own data.

9 Groups

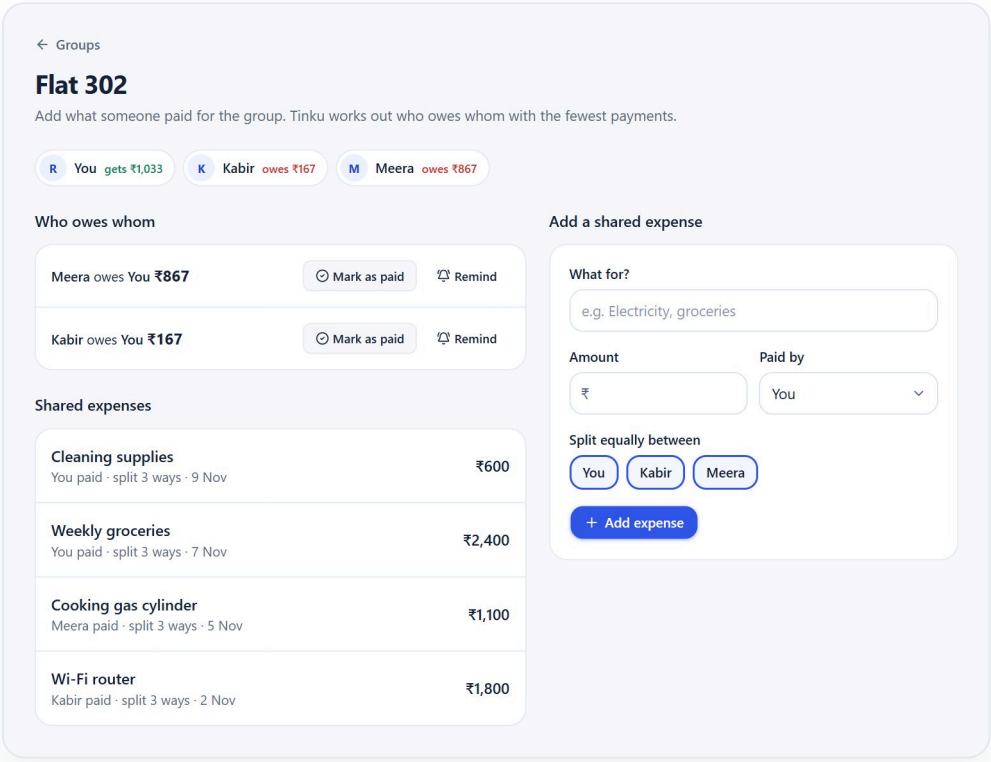
Menu → Together → Groups

`/groups` `/groups/new` `/groups/[id]`

Share costs with **flatmates, family or friends**, and always know who has paid and who still owes.



Groups list: members, number of shared expenses, and your balance in each group.



A group: member balances, who owes whom (fewest payments), shared expenses, and the add form.

Flow

1. **New group**
   Name it and list the other people; you're added automatically.
2. **Add a shared expense**
   What for · amount · paid by · split equally between (tap people). Shows each person's share.
3. **See who owes whom**
   Balances are simplified to the fewest payments.
4. **Settle or remind**
   "Mark as paid" records a settlement; "Remind" sends a reminder.

Behind the scenes

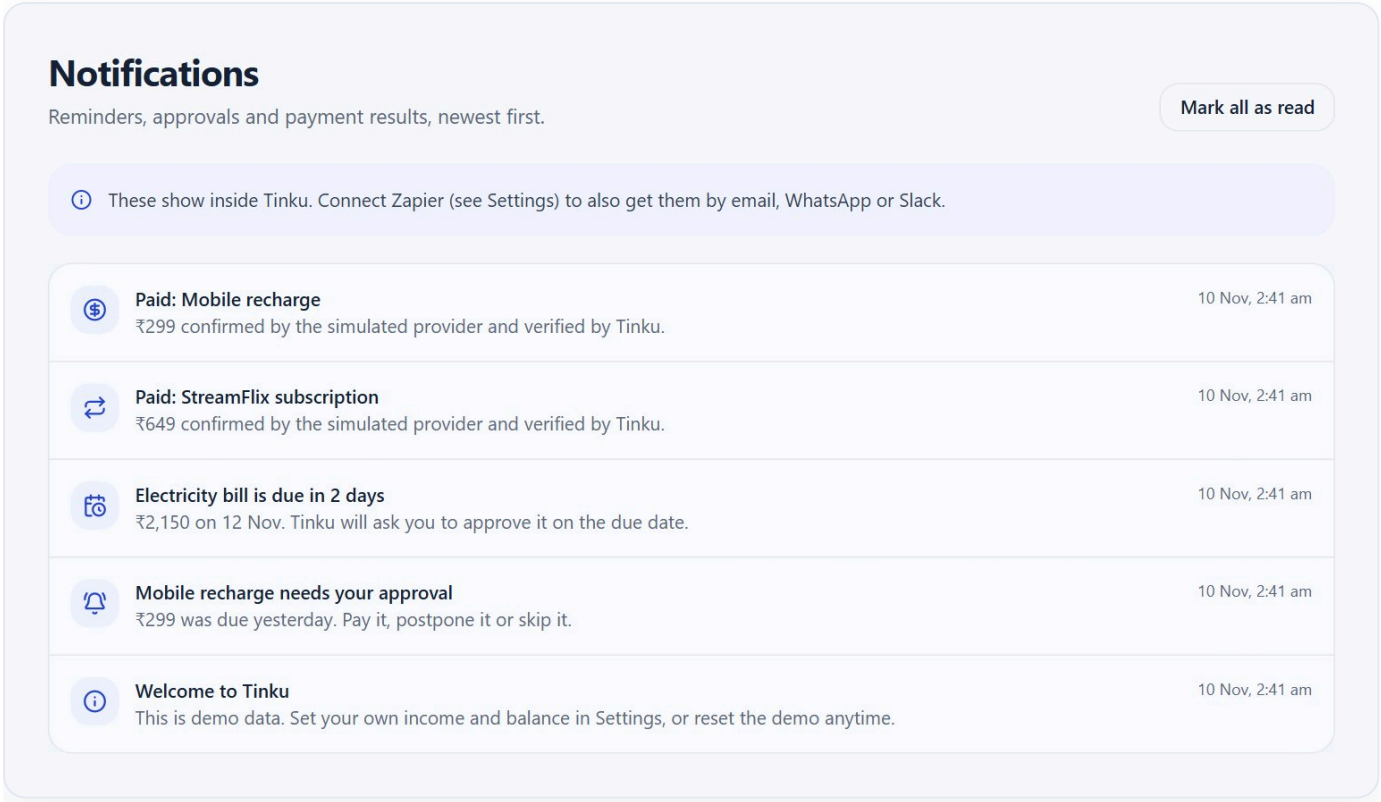
- Member chips show "gets ₹X", "owes ₹X" or "settled".
- Reminders become notifications. With Zapier connected, they go out by email or WhatsApp.
- Used by the Crisis drill's "Ask family or flatmates" step.

10 Notifications

Bell in the top bar (shows the unread count)

`/notifications`

One inbox for **reminders, approvals and payment results**, newest first.



Each notification links to the page where you can act on it.

Flow

1. **See the bell count**
   Visible on every page.
2. **Open Notifications**
   Unread items are highlighted.
3. **Tap one**
   Jumps to the bill, payment, drill or group it's about.
4. **Mark all as read**
   Clears the count.

What creates a notification

Reminder	3 days before a due date; on the day for reminder bills
Approval	An "Ask before paying" bill is due
Payment	Verified, pending, failed, refunded, marked paid
AutoPay	Paid, blocked (with the reason), turned on, cancelled
Drill · Group · System	Drill saved · settle-ups and reminders · welcome

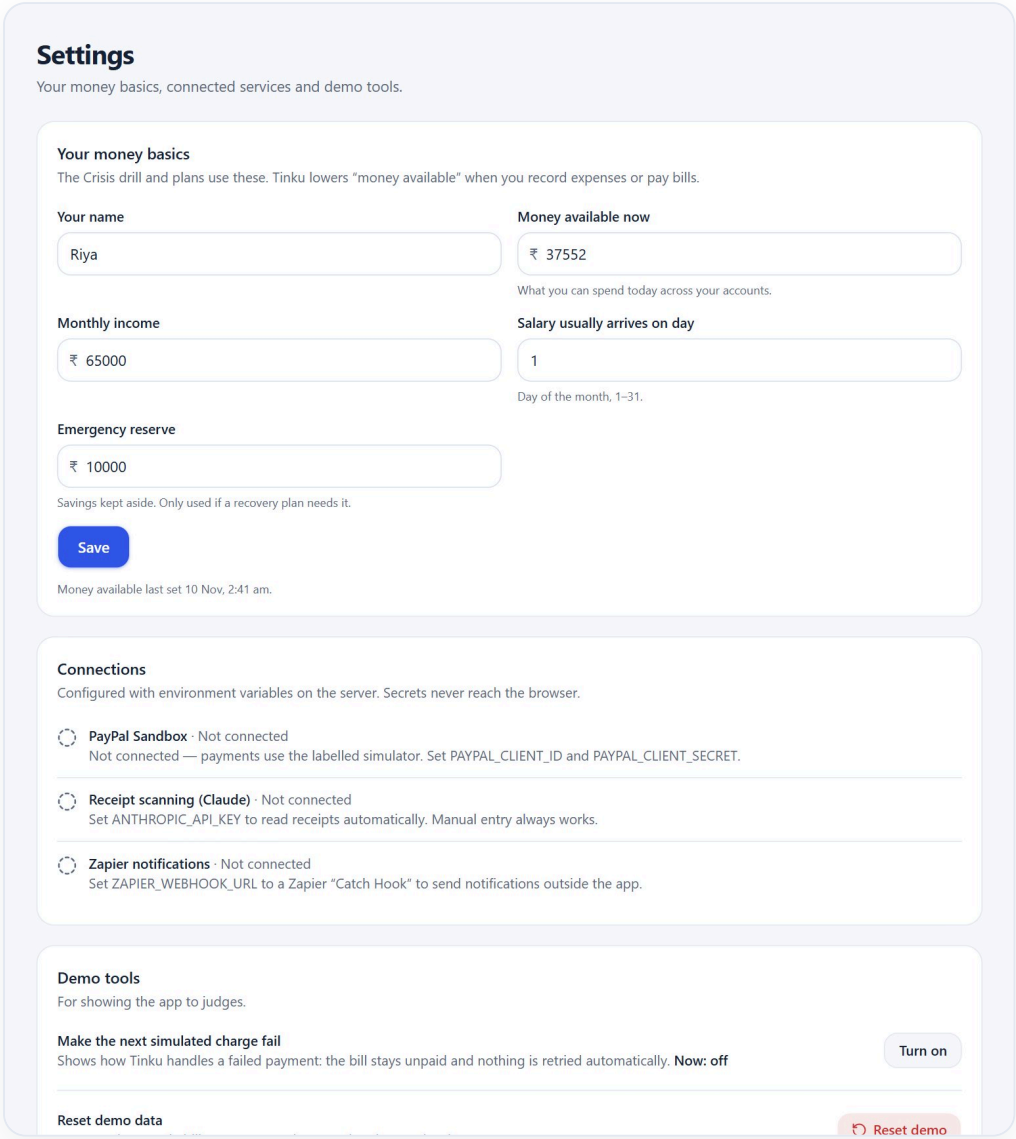
Zapier: with `ZAPIER_WEBHOOK_URL` set, every notification is also sent to a Zap (email, WhatsApp, Slack, Google Sheets). Zapier only delivers messages; it never decides payments.

11 Settings

Bottom of the menu

`/settings`

Your **money basics**, which services are **connected**, and **demo tools** for presenting.



Money basics, connections and demo tools.

Flow

1. **Set your money basics**
   Name · money available now · monthly income · salary day · emergency reserve → Save.
2. **Check connections**
   PayPal Sandbox · receipt scanning (Claude) · Zapier, each showing Connected or which key to add.
3. **Use demo tools**
   "Make the next simulated charge fail" shows failure handling; "Reset demo data" restores the sample.

Where these values are used

- Home tiles (money available).
- Crisis drill: starting balance, salary amount and day, reserve.
- Keys are set as server environment variables and never shown in the browser.

A End-to-end journeys & statuses

How the pages connect

Journey 1 · An “Ask before paying” bill from due date to verified payment

Daily check — Bill due → “Needs approval” + notification
› **Home / Bills** — Pay now
› **Review** — ₹ amount, USD demo charge
› **Approve** — PayPal Sandbox or simulator
› **Server verifies** — Status, amount, currency, reference
› **Result** — Paid · expense · next due · history

Journey 2 · An AutoPay month

Add bill — Mode: AutoPay
› **Authorize** — Limit, months, consent
› **3 days before** — Reminder notification
› **Due date** — Checks pass → charge
› **Notified** — Verified, or blocked/failed with a reason

Journey 3 · From an emergency rehearsal to getting help

Crisis drill — Pick “Salary 30 days late”
› **Verdict** — Short by ₹X, bills at risk
› **Plan** — Reserve → trim → pause → ask
› **Save** — Drill history + notification
› **Groups** — Request & track contributions

A bill payment’s status

Upcoming → due date → Needs you → you pay → Processing → Paid

“Needs you” shows as *Needs approval* (Ask before paying), *Due* (Remind me) or *Needs you* (AutoPay that was blocked). Other endings: Paid outside app · Skipped (unpaid) · Payment failed → Try again. Postpone moves just this one back to Upcoming. Paid, paid outside and skipped close it and open the next one for repeating bills.

A payment attempt’s status

Started → ends as one of → Verified · Pending · Failed · Cancelled

Pending is re-checked with PayPal until it becomes Verified or Failed. A Verified payment never goes back; it can only become Refunded, and only when the provider reports a refund.

Page routes

Home · Expenses · Add expense	`/` · `/expenses` · `/expenses/new`
Budgets · Edit limits	`/budgets` · `/budgets/edit`
Bills · Add · Details · Edit	`/bills` · `/bills/new` · `/bills/[id]` · `/bills/[id]/edit`
Pay · Simulated checkout · Result	`/pay/[occurrenceId]` · `/pay/simulated` · `/payments/[id]`
AutoPay · Authorize · History	`/autopay` · `/autopay/new?bill=…` · `/history`
Crisis drill · Groups · Notifications · Settings	`/drill` · `/groups` (`/new`, `/[id]`) · `/notifications` · `/settings`

Server endpoints (for the tech team)

`GET /api/paypal/return`	PayPal sends the buyer back here; the server captures, verifies and records the payment.
`GET /api/paypal/cancel`	Buyer cancelled at PayPal; the payment is marked cancelled and nothing is charged.
`POST /api/webhooks/paypal`	Signature-verified PAYMENT.CAPTURE.* events, de-duplicated by event id.
`POST /api/scheduler/run`	Daily check for a cron job (Bearer CRON_SECRET): reminders, due bills, AutoPay.
`POST /api/receipts/extract`	Receipt photo/PDF → Claude → suggested merchant, total, date, category.